Class-10

Elasticity of Demand

(i) Price elasticity of Demand

(ii) Income elasticity of Demand

(iii) Cross elasticity of Demand

→ Price elasticity of Demand

↳ shows the responsiveness of percentage change in demand due to the percentage change in price.

$$E_P / E_d = \frac{\text{Percentage change in } Q_d}{\text{Percentage change in Price}}$$

$$= \frac{\Delta Q / Q \times 100}{\Delta P / P \times 100}$$

$$= \frac{\Delta Q}{Q} \times \frac{P}{\Delta P}$$

$$\boxed{\therefore E_P = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}}$$; EP = Price elasticity of demand

ΔQ = change in demand
$= Q_2 - Q_1$; Q_2 = Current demand
Q_1 = initial demand

ΔP = change in Price
$= P_2 - P_1$

Math-1: Suppose the ~~#~~ price of a product increased from Tk 50 to Tk 75 and its demand has decreased by 100 units to 95 units.

(i) Calculate the Price elasticity of demand.

(ii) Find out whether the demand for this good is elastic or inelastic.

$\Delta Qd > \Delta P$ $\qquad$ $\Delta Qd < \Delta P$

(i)

Given,

Initial Price (P_1) = Tk 50

Current Price (P_2) = Tk 75

Initial Demand (Q_1) = 100 units

Current Demand (Q_2) = 95 unit

Now,

$$EP = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$= \frac{95-100}{75-50} \times \frac{50}{100}$$

$$= -0.1$$

Interpretation:

If price increases from TK50 to Tk75 demand has decreased by 0.1 Percent.

(ii)

From the above calculation, $E_P = -0.1$

Setting the value in modulas we get,

$|E_P| = |-0.1| = 0.1$ which is less than 1.

Rules,

If, $|E_P| > 1 \rightarrow$ the demand is elastic

If, $|E_P| < 1 \rightarrow$ the demand is inelastic

If, $|E_P| = 1 \rightarrow$ the demand is unit elastic

Therefore, in question the demand for this good is inelastic.

Income Elasticity of demand

$$E_Y = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

$Y\uparrow D\uparrow$ (+) Normal goods

$Y\uparrow D\downarrow$ Inferior good (−)

Suppose, the income of a person increased from Tk 50000 to Tk 70000 and his demand has decreased by 100 unit to 95 unit.

(i) Calculate the income elasticity of demand

(ii) Find out whether the good in question is a normal or inferior good.

Given,

Initial income $(Y_1) = 50000$ Tk

Current " $(Y_2) = 70000$ Tk

Initial Demand $(Q_1) = 100$ unit

Current " $(Q_2) = 95$ unit

Now,

$$E_Y = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

$$= \frac{95-100}{70000-50000} \times \frac{50000}{100}$$

$$\therefore E_Y = -0.125$$

Interpretation:

if income increased from Tk 50000 to Tk 70000 demand has decreased by 0.125 Percent.

(ii)

From the above calculation $E_Y = -0.125$, which is a negative number.

Rules:

if E_Y is Positive the good is normal good.

if E_Y is negative the good is inferior good.

erefore, the good given in question is an inferior good.

Cross Elasticity of demand

$$E_C = \frac{\Delta Q_{coffee}}{\Delta P_{Tea}} \times \frac{P_{Tea}}{Q_{coffee}}$$

Tea Coffee (+) → Substitutes
P↑ D↑
Milk Coffee (−) → Complement
P↑ D↓

▣ Suppose the price of Teherin increased from Tk 150/Pack to Tk 300/Pack and as a result, the demand of Kacchi has increased from 300 units to 500 units.

(i) Calculate the Cross elasticity of demand.

(ii) Find out whether the inter-related goods in question are substitutes or complement.

(i) Given,

Initial Price of Tehari (P_1) = Tk 150/pack
Current ,, ,, ,, (P_2) = Tk 300/Pack
Initial demand of kacchi (Q_1) = 300 units
Current ,, ,, ,, (Q_2) = 500 units

We know,

$$E_C = \frac{\Delta Q_{kacchi}}{\Delta P_{Tehari}} \times \frac{P_{Tehari}}{Q_{kacchi}}$$

$$= \frac{500-300}{300-150} \times \frac{150}{300}$$

$$= 0.67$$

Interpretation:

If the price of Tehari increased from Tk 150/pack to Tk 300/pack the daemand for kacchi increased 0.67 Percent.

(ii)

From the above calculation $EC = 0.67$ which is a positive number.

Rules:

if EC is positive the inter-related goods are substitute.
if EC is negative the inter-related goods are complement.

Therefore the inter-related goods in question are substitute.